Expermint 3 -Write the Python Program for Water Jug Problem

Aim:

To write a python program for water jug problem.

Algorithm:

Step1: Start with two empty jugs of different capacities, and a target quantity of water to measure.

Step2: Fill one jug to its maximum capacity.

Step3: Pour the water from the full jug into the other jug until it is full or the first jug is empty.

Step4: If the second jug is full, pour out the water to empty it.

Step5: Repeat steps 2-4, filling and pouring water between the two jugs, until the desired quantity of water is measured or all possible combinations are tried.

Program:

```
from collections import deque

def water_jug_problem(x, y, z):
    """
    Solves the Water Jug Problem using Breadth-First Search (BFS).

    x = capacity of Jug 1
    y = capacity of Jug 2
    z = target amount of water
    """

    queue = deque([(0, 0, [])])  # Initial state
    visited = set()

    while queue:

        a, b, steps = queue.popleft()

        if (a, b) in visited:
            continue

        visited.add((a, b))
```

```
        # Goal Test
        if a == z or b == z:
            return steps, (a, b)

        # Fill Jug 1
        queue.append((x, b, steps + ["Fill Jug 1"]))

        # Fill Jug 2
        queue.append((a, y, steps + ["Fill Jug 2"]))

        # Empty Jug 1
        queue.append((0, b, steps + ["Empty Jug 1"]))

        # Empty Jug 2
        queue.append((a, 0, steps + ["Empty Jug 2"]))

        # Pour Jug 1 into Jug 2
        amount = min(a, y - b)
        queue.append(
            (a - amount, b + amount,
             steps + ["Pour Jug 1 into Jug 2"])
        )

        # Pour Jug 2 into Jug 1
        amount = min(x - a, b)
        queue.append(
            (a + amount, b - amount,
             steps + ["Pour Jug 2 into Jug 1"])
        )

    return None


result = water_jug_problem(4, 3, 2)

if result:
    steps, state = result

    print("Steps:")
    print("\n".join(steps))

    print("\nGoal State:", state)

else:
print("No solution found.")
```

Output:

```
Steps:
fill jug 2
pour jug 2 into jug 1
fill jug 2
pour jug 2 into jug 1

Goal State: (4, 2)
```

Result: Hence, the simple python program for water jug problem is done